

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 13

POSITIVE OR NEGATIVE

AIM: To write and execute an 8085 assembly language program to find positive or negative and store the result in memory.

ALGORITHM

1. Load the number into accumulator A.
2. Check the most significant bit (MSB) using ANI 80H.
3. If the result is zero, the number is positive.
4. If the result is non-zero, the number is negative.
5. Store the result.
6. Stop the program.

CODE:

```
MVI A,25H  
ANI 80H  
JNZ NEGATIVE  
MVI A,00H  
STA 8050H  
HLT  
NEGATIVE: MVI A,01H  
STA 8050H  
HLT
```

OUTPUT:

FileResetAssemblerDebugHelp

Registers

A01

BC0000

DE0000

HL0000

PSW0000

PC420D

SPFFFF

Int-Reg00

Flag

S0

Z0

AC0

P0

C0

Decimal - Hex Conversion

Decimal

0

Hex

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

2500

-

+

0B

Update Memory

Load me at

1LDA 2500

2ORA A

3JM NEGATIVE

4

5MVI A, 01

6STA 2501

7HLT

8

9NEGATIVE: MVI A, 00

10STA 2501

11HLT

DataStackKeyPadMemoryI/O Ports

Start2500

OK

Address (Hex)AddressData

09C4250011

09C525011

09C625020

09C725030

09C825040

09C925050

09CA25060

09CB25070

09CC25080

09CD25090

09CE25100

09CF25110

09D025120

09D125130

Line NoAssembler Message

0

Program assembled successfully

Simulator: Idle

Search

ENG

IN

11:51

28-07-2026